TRAINEX | ESP32 Wiring & GPIO Mapping

Revision: 8 November 2026 | **Hardware:** 30-pin classic ESP32-WROOM-32, AS5600 + five core controls

1. Core GPIO allocation

Control / signal	ESP32 pin	Wiring / behavior
AS5600 I2C SDA	GPIO21	To SDA on 3.3 V sensor board
AS5600 I2C SCL	GPIO22	To SCL on sensor board
AS5600 supply	3V3 + GND	Sensor VCC to 3.3 V, GND common
Vigilance acknowledge	GPIO23	Normally-open pushbutton to GND
Emergency Brake RELEASE	GPIO25	Normally-open pushbutton to GND
Door OPEN	GPIO26	Normally-open pushbutton to GND
Door CLOSE	GPIO27	Normally-open pushbutton to GND
HORN	GPIO33	Normally-open pushbutton to GND
Buzzer driver OUTPUT	GPIO32	Optional NPN transistor only; default disabled

Optional extra buttons: GPIO18 LEFT ENABLE, GPIO19 RIGHT ENABLE, GPIO13 SECURITY BRAKE. Set TRAINEX_ENABLE_EXTRA_CONTROLS=1 only after installing them.

2. How to wire the ground rail

Run one ESP32 GND lead to the breadboard blue negative rail. Bridge any break in that rail with a jumper and confirm continuity using a multimeter. Each button has one terminal on its GPIO and the other on this common GND rail. Connect the AS5600 GND to the same rail. Do not bridge the center IC trench or connect the buttons to 5 V.

3. Master-handle calibration

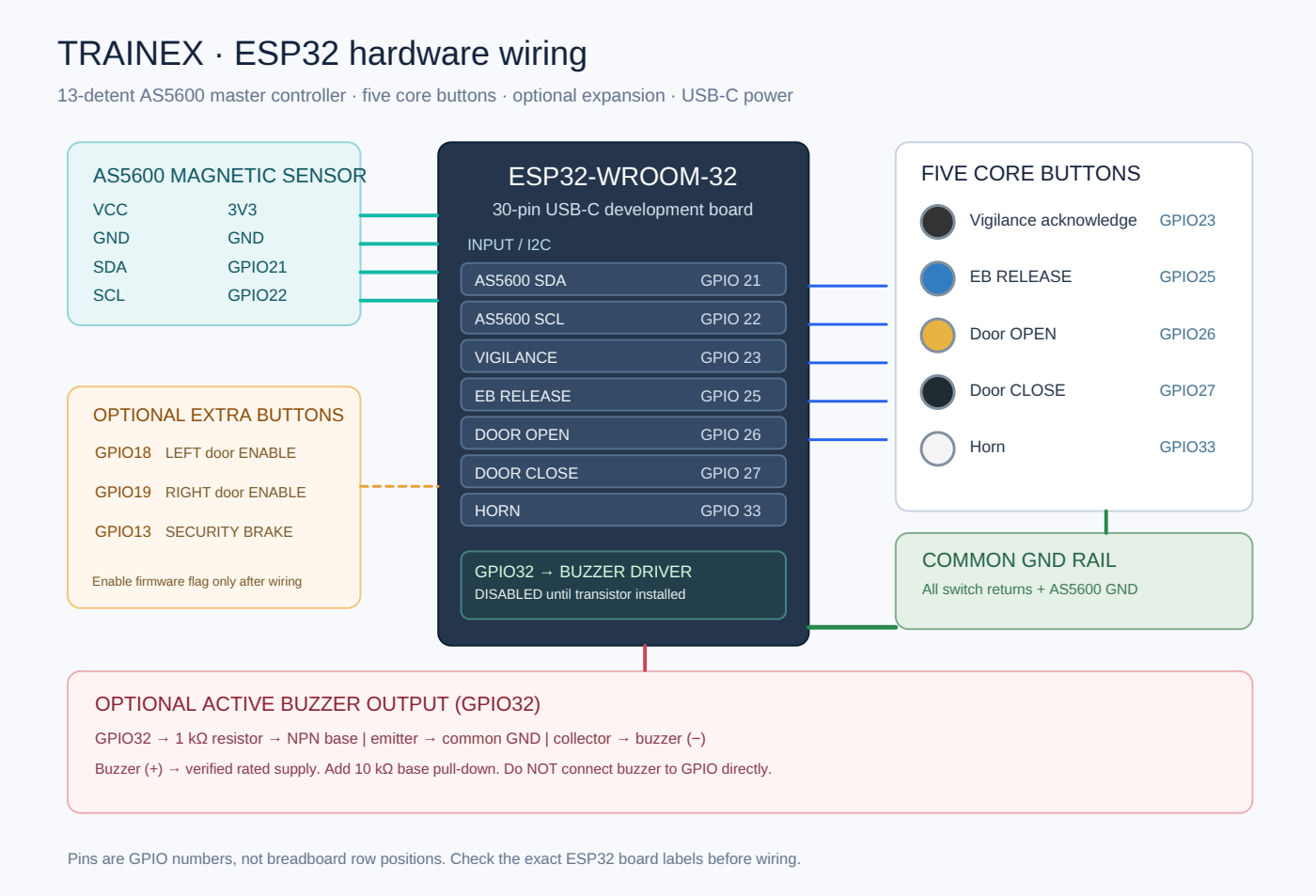
The former 8-contact master selector is retired. Disconnect its contacts, especially from GPIO21/22. Install the shaft magnet and sensor. With handle at N, open PlatformIO Serial Monitor (115200), read AS5600 RAW, and enter the value (0–4095) in include/TrainexHardwareConfig.h. Default -1 prevents uncalibrated master commands. Check P1 direction, set REVERSE=1 if needed, then check all 13 notches.

4. Emergency Brake release

Physical GPIO25 sends EMERGENCY_BRAKE_RELEASE, not RESET. The website engine accepts release only at standstill with the master handle at N; the simulated release takes 3 seconds. The master handle EB detent still activates emergency braking. The red SECURITY BRAKE button is optional additional hardware.

TRAINEX | Wiring Reference Diagram

Conceptual connection map. Pin labels refer to ESP32 GPIO numbers; this is not a breadboard hole-by-hole placement drawing.



Optional active buzzer: the PDF lists a 3–24 V active piezo. Connect GPIO32 through a transistor base resistor (suggested 1kΩ), add a 10kΩ base pull-down, and power the buzzer from a verified correctly rated supply. Join grounds. Enable the driver flag only after checking load current.

Wi-Fi connection: USB-C is used to program and diagnose the board; the Vercel website needs a separately deployed authenticated WSS relay and an active paired TRAINEX session. The ESP32 does not run train physics or vigilance timing.

5. Final verification checklist

Verify: AS5600 detected; all 13 detents; GPIO23 vigilance; GPIO25 EB release; GPIO26 OPEN; GPIO27 CLOSE; GPIO33 horn; no phantom inputs; door side enabled before opening; correct All Doors Closed state; buzzer driver installed or disabled. Verify split-rail continuity and check for shorts before powering.

Source: User-provided *Hardware Materials & Technical Specifications.pdf* (6 pages). Extra switches and buzzer-driver parts are explicit proposed additions, not part of the original materials list.